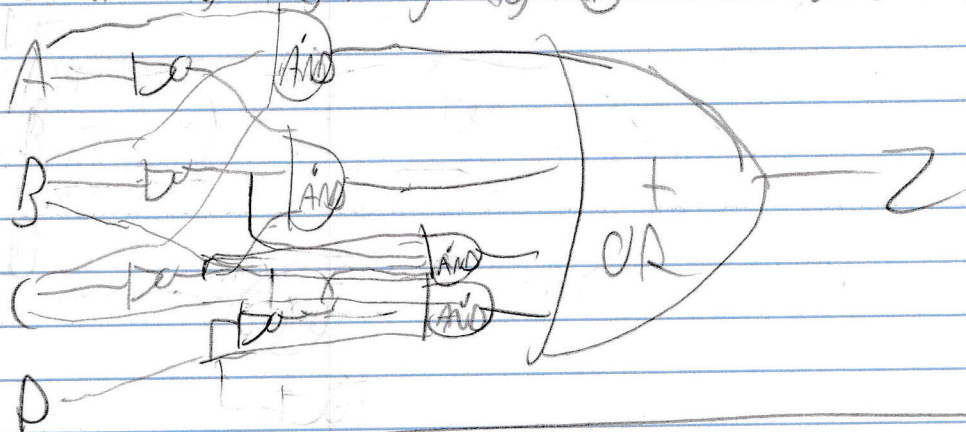


4.13 HW #4

combinations, $ABC, \overline{A}BC, B\overline{C}D, B\overline{C}\overline{D} \rightarrow (ABC) + (\overline{A}BC) + (B\overline{C}D) + (B\overline{C}\overline{D})$



4.16

A) $\Sigma m(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15) = Z$

$\Sigma m(4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15) = Y_1$

$\Sigma d(w) = \text{don't care}$

$\Sigma m(2, 3, 8, 9, 10, 11, 12, 13, 14, 15) = Y_0$

$\Sigma d(w) = \text{don't care}$

B) $\Pi M(w) = Z$

$\Pi M(1, 2, 3) = Y_1$

$\Pi d(w) = \text{don't care}$

$\Pi M(1, 4, 5, 6, 7) = Y_2$

$\Pi d(w) = \text{don't care}$

x_3	x_2	x_1	x_0	Z	y_1	y_0
0	0	0	0	0	0	0
0	0	0	1	1	0	0
0	0	1	0	1	0	1
0	0	1	1	1	0	1
0	1	0	0	1	1	0
0	1	0	1	1	1	0
0	1	1	0	1	1	0
0	1	1	1	1	1	0
1	0	0	0	1	1	1
1	0	0	1	1	1	1
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	1	1	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

4.19

A) $Z = (AB) + \overline{C}, Z = A(B + \overline{C})$

C) $D = (VL) + S, D = V(L + S)$

4.25

A) $\Sigma m(5, 6, 7, 10, 11, 13, 14, 15)$

$\Pi M(0, 1, 2, 3, 4, 8, 9, 12)$

B) $\Sigma m(0, 2, 4, 6)$

$\Pi M(1, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15)$

C) $\Sigma m(2, 11, 13, 14, 15)$

$\Pi M(0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 12)$

D) $\Sigma m(4, 8, 12, 13, 14)$

$\Pi M(0, 1, 2, 3, 5, 6, 7, 9, 10, 11, 15)$

A	B	C	D	E	F	G	H	I	J
0	0	0	0	0	0	1	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0
0	0	1	1	0	0	0	0	0	0
0	1	0	0	0	0	1	0	0	0
0	1	0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0

4.27

A) $\Sigma m(4,6,7)$

B) $\Pi M(0,1,2,3,5)$

A	B	C	A(B+C)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

4.32

C) $\Sigma m(3,4,5,7)$

$\Sigma d(2,6)$

D) $\Sigma(0,3,4)$

$\Sigma d(1,2,5,6,7)$

$ABC + A\bar{B}C + AB\bar{C} + A\bar{B}\bar{C} \rightarrow B(\bar{A}A) + A\bar{B}C + AB\bar{C}$

$BC + AB(\bar{C} + C)$

$(A+BC)(B+C)$

$(A+BC)(B+C)$

$AB + AC + 0 + BC \rightarrow AB + BC$

$\bar{A}BC + \bar{A}B\bar{C} + A\bar{B}\bar{C}$

$\bar{B}C(\bar{A} + A)$

$\bar{B}C + \bar{A}BC + A\bar{B}\bar{C}$

$\bar{B}C + BC(\bar{A} + A)$

$\bar{B}C + BC$

4.37

A) $A B C D$

$0 0 0 0$

$0 0 0 1$

$0 0 1 0$

$0 0 1 1$

$0 1 0 0$

$0 1 0 1$

$0 1 1 0$

$0 1 1 1$

$1 0 0 0$

$1 0 0 1$

N

0

1

2

3

4

5

6

7

8

9

$5 \cdot N$

0

5

10

15

20

25

30

35

40

45

$S T U V$

$0 0 0 0$

$0 0 0 1$

$0 0 1 0$

$0 0 1 1$

$0 1 0 0$

$0 1 0 1$

$0 1 1 0$

$0 1 1 1$

$1 0 0 0$

$1 0 0 1$

$W X Y Z$

$0 0 0 0$

$0 1 0 1$

$0 0 0 0$

$0 1 0 1$

$0 0 0 0$

$0 1 0 1$

$0 0 0 0$

$0 1 0 1$

$0 0 0 0$

$0 1 0 1$

B) $S=0, T=A, U=B, V=C$
 $W=0, X=B, Y=0, Z=D$

X	Y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

